Figure 2.8 An example of a blueprint of my home, showing dimensions and material specifications that are an important source of spatial information when modelling physical structures digitally.

ENGINEERING DRAWINGS

Engineering drawings are the definitive expression of engineering intent, providing a universal standard that uses symbols, conventions, and annotations to define an object's design in precise detail. They provide essential information throughout the object's lifecycle.

The typical layered structure of these drawings supports selective information extraction. Layers for dimensions, annotations, and geometric features can be processed independently to isolate data specific to modeling needs.

Crucially, drawing revision systems provide historical tracking of design changes. Understanding this design evolution ensures that the digital representation matches the system's current configuration, accounting for field modifications or design improvements.

The engineering drawing in figure 2.9 shows the geometry and dimensions of an air conditioning unit I have installed in my home. The drawing conveys the geometry and dimensions of the physical object in a multi-view projection. This information can be used both to plan the placement of the physical device as well as to create an accurate 3D model of the device to be placed in a 3D visualization.